

Name: _____

Double Award Science – Chemistry Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Scores

Year	G	F	E	D	C	Max UMS
2017	10	13	16	19	23	27
2018	9	13	17	21	26	31
2019	9	14	19	24	29	34
2022	8	11	14	17	21	26
2023	13	17	21	25	29	33

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – **NEW**

3430U20-1



S17-3430U20-1

SCIENCE (Double Award)**Unit 2: CHEMISTRY 1
FOUNDATION TIER**

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	4	
3.	10	
4.	12	
5.	7	
6.	6	
7.	9	
8.	6	
Total	60	

3430U201
01**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

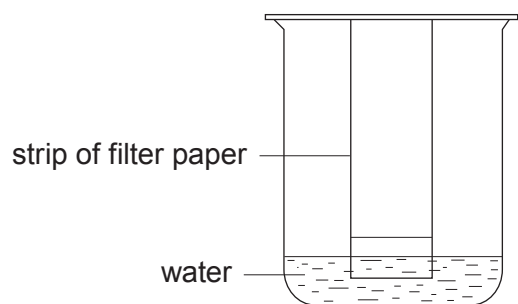
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



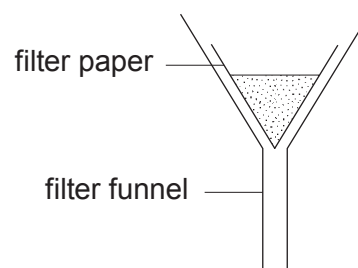
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Answer all questions.

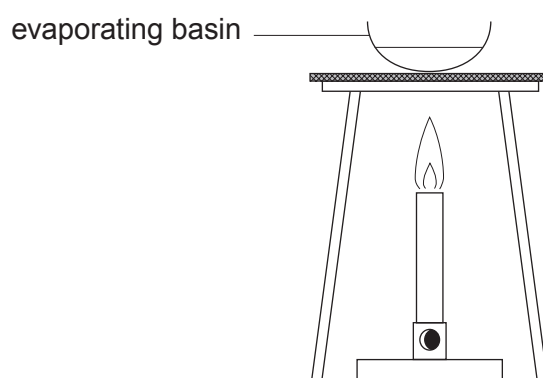
1. (a) The diagrams show three methods, **A**, **B** and **C**, used to separate mixtures.



A



B



C

Give the letter, **A**, **B** or **C**, of the method you would use to:

- (i) remove sand from sea-water,

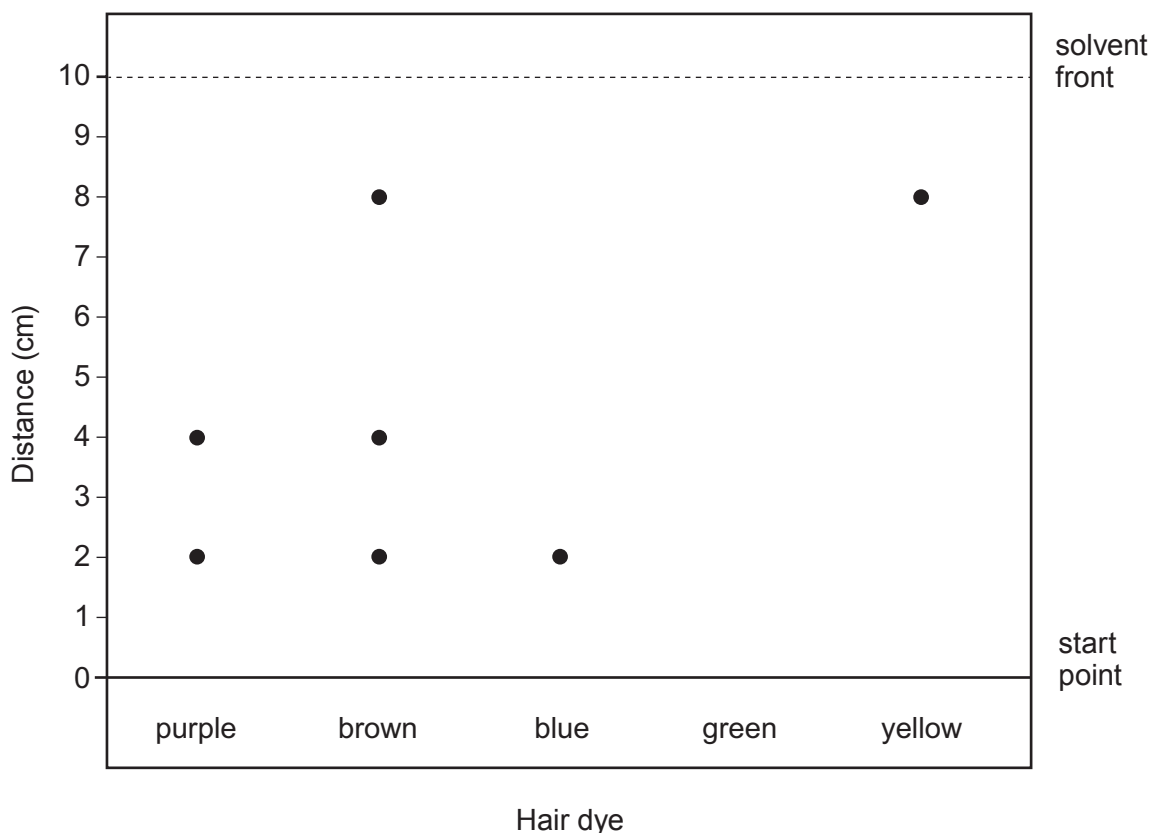
[1]

- (ii) show that sea-water contains salt.

[1]



(b) Paper chromatography was used to show the pigments present in different hair dyes.



(i) Give the **two** dyes that are mixed to make brown hair dye. [1]

..... and

(ii) Green hair dye is made by mixing together blue and yellow dyes.
Draw on the diagram the result that you would expect to see for green hair dye. [1]

(iii) The R_f value of a substance can be used to identify that substance.
The R_f value of the pigment in red hair dye is 0.4.

Use the equation below to calculate the distance this red hair dye would have moved on the diagram. [2]

$$\text{distance moved} = R_f \times \text{distance moved by solvent}$$

distance moved = cm

6



2. The chemical formula of potassium carbonate is K_2CO_3 .

(a) (i) State how many carbon atoms are present in the formula, K_2CO_3 [1]

(ii) Give the **total** number of atoms shown in the formula, K_2CO_3 [1]

(b) Calculate the percentage by mass of carbon in potassium carbonate, K_2CO_3 . [2]

percentage by mass = %



3. (a) Most of the tap water in Wales comes from mountain reservoirs. The water from these reservoirs is never pure.

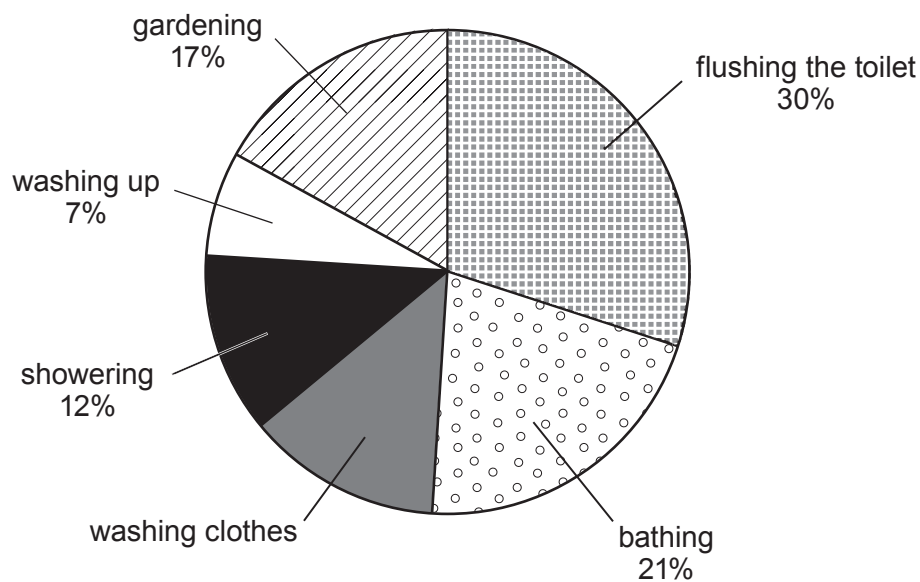
- (i) Information about some of the stages used in the treatment of tap water is shown.

Draw a line to connect each stage to the correct description. One has already been completed. [2]

Stage	Description
filtration	large insoluble particles are allowed to settle out as sludge
screening	screens are used to remove objects such as leaves, sticks and litter
chlorination	harmful bacteria are killed to make the water safe to drink
sedimentation	small insoluble particles are removed by passing the water through sand beds



(ii) The pie chart shows some of the main uses of water in a typical Welsh home.



Use the pie chart to answer parts I and II.

I. Calculate the percentage of water used for washing clothes. [1]

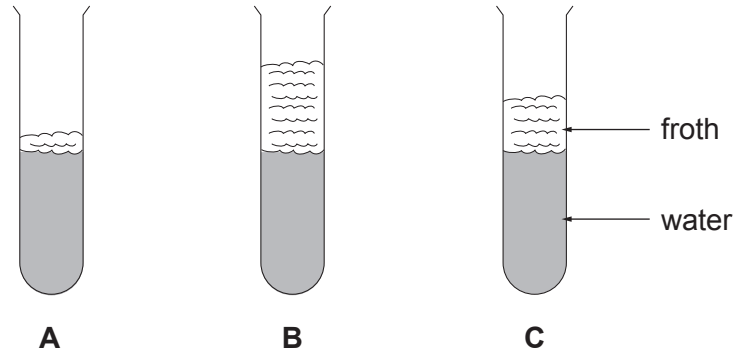
..... %

II. Give the name of the use that accounts for three times as much water as another use. [1]

.....



- (b) 5 cm³ of three water samples, labelled **A**, **B** and **C**, were mixed with 10 drops of soap solution and shaken. One of the samples was distilled water. The boiling point of each water sample was also measured. The results obtained are shown.



Water sample	Boiling point (°C)
A	102
B	100
C	101

- (i) Place the water samples, **A**, **B** and **C**, in order of hardness.

[1]

hardest

.....

softest

- (ii) Give the letter of the sample that is distilled water. Give **two** reasons for your choice. [2]

Sample

Reasons

1.

.....

2.

.....



- (iii) The volume of each water sample and soap solution was kept the same for each test. State which other variable must also be controlled before measuring the froth height. Describe how you would control this. [2]

.....

.....

.....

- (iv) Hardness in water can be caused by dissolved calcium fluoride. Give the chemical formula for calcium fluoride. [1]

.....

10



4. When magnesium ribbon is added to excess hydrochloric acid, HCl, magnesium chloride and hydrogen gas are produced.

magnesium + hydrochloric acid $\longrightarrow$ magnesium chloride + hydrogen

- (a) Suggest **two** things that you would expect to **see** during this reaction. [2]

1.

2.

- (b) Write the **symbol** equation for the reaction that takes place. [3]

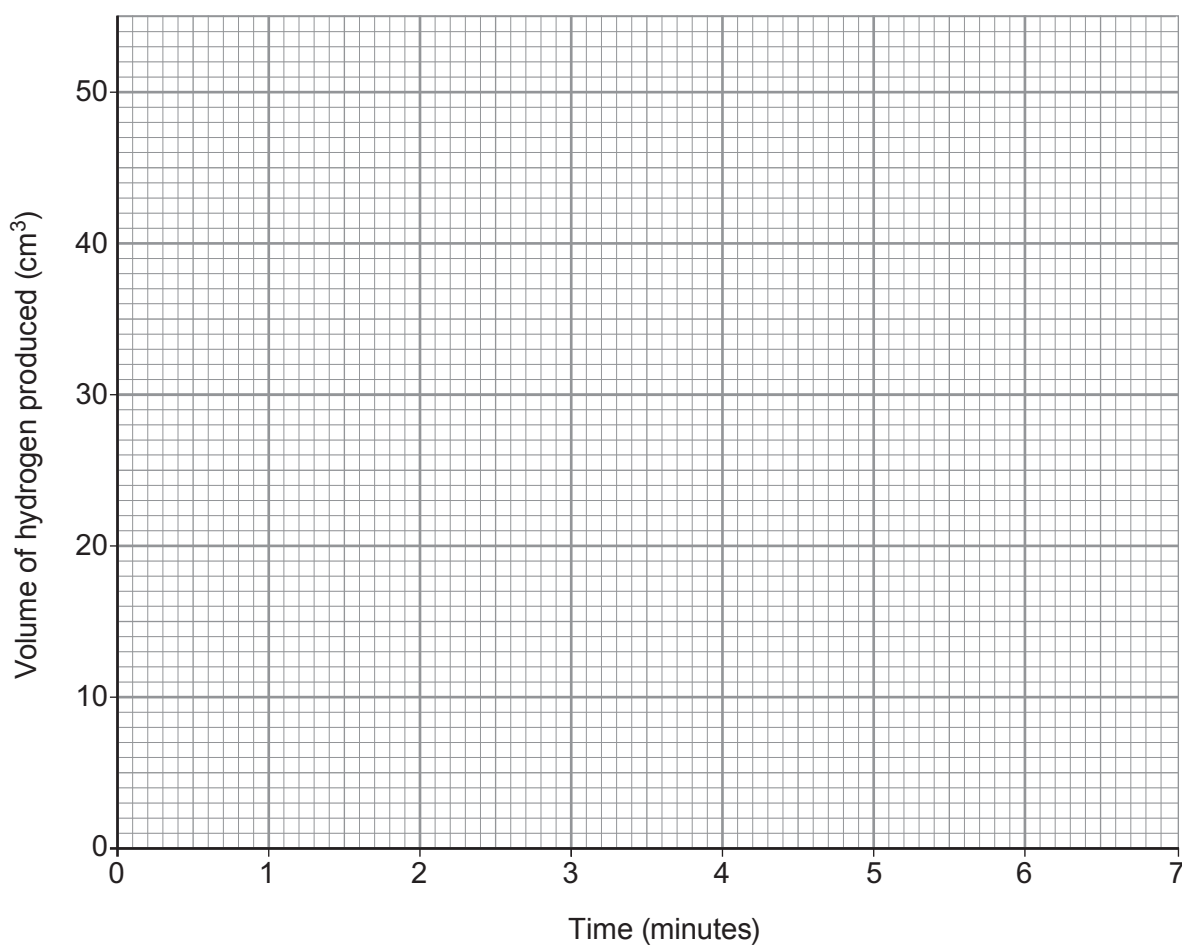
..... + $\longrightarrow$ +

- (c) Amelia recorded the volume of gas produced for 7 minutes for this reaction. She carried out the reaction at room temperature, 20 °C. The results she collected are shown in the table.

Time (minutes)	Volume of hydrogen produced (cm ³)
0	0
1	15
2	28
3	32
4	45
5	49
6	50
7	50

- (i) Plot a graph of her results **on the grid opposite**. Draw a suitable line and label it **A**. [3]





(ii) Use the graph to find:

I. the volume of gas produced in 90 seconds, [1]

II. the time taken for the reaction to end. [1]

(iii) **On the same grid**, sketch the graph that would be obtained if Amelia repeated the same experiment at 40 °C. Label it **B**. [2]



5. The following information is about the Group 7 elements of the Periodic Table, called the halogens.

Group 7 – the halogens

Group 7 – the halogens																		He
H												B	C	N	O	F	Ne	
Li	Be												Al	Si	P	S	Cl	Ar
Na	Mg											Ga	Ge	As	Se	Br	Kr	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	In	Sn	Sb	Te	I	Xe	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Ti	Pb	Bi	Po	At	Rn	
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg							
Fr	Ra																	

The Group 7 elements all exist as diatomic molecules, meaning their atoms go around in pairs. Fluorine is found at the top of the group, astatine is at the bottom of the group.

Chlorine, atomic number 17, is a pale green gas at room temperature (20°C). It has a melting point of -101°C. Bromine, atomic number 35, is a red-brown liquid at room temperature. It has a melting point of -7°C. Iodine, atomic number 53, is a grey solid at room temperature. It has a melting point of 114°C.

Chlorine is used to make bleaches, to treat public drinking water and to sterilise public swimming pools. Bromine is used by gardeners in pesticides, to prevent insects from harming plants. Iodine solution is used as an antiseptic on wounds and on patients before being operated on.

- (a) Which **one** of these statements best describes the trend in the melting points of the Group 7 elements as you go down the group? Tick (✓) the correct answer. [1]

The melting point increases

☐

The melting point decreases

☐

The melting point increases and then decreases

☐

There is no trend in the melting point

☐

- (b) What is the correct formula for chlorine gas? Tick (✓) the correct answer. [1]

CL2

☐

Cl₂

☐

cl₂

☐

2Cl

☐


- (c) Predict a value for the boiling point of chlorine and explain your answer. [2]

Boiling point of chlorine °C

Explanation

- (d) Give the property that the Group 7 elements have that means they can be used as described in the text. [1]

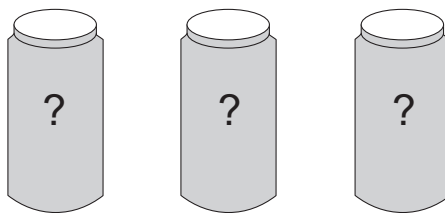
- (e) Decide whether the following statements about astatine are true or false. [2]

Astatine ...	True / False
... is a solid at room temperature	
... conducts electricity	
... has a melting point higher than 114 °C	
... is coloured	

7



6. You have been given three jars containing the compounds **potassium bromide**, **barium chloride** and **calcium iodide**.



Unfortunately the labels have been removed from the jars and, because they are all white compounds, they cannot be identified by their appearance.

Describe **two** tests that you should carry out **on all three** of the compounds, so that each compound can be identified.

State how you would carry out both tests and give the expected observations for each compound. [6 QER]

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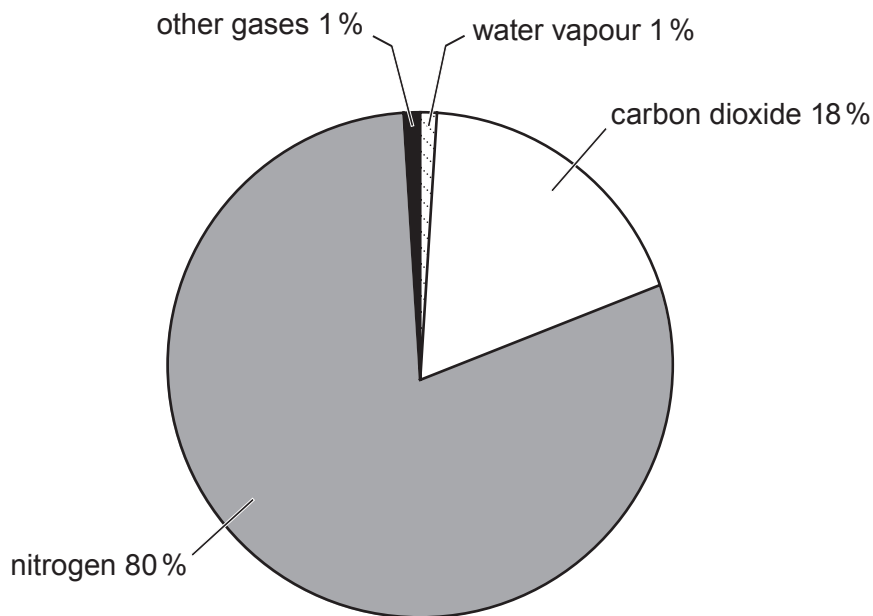
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7. (a) The Earth’s atmosphere has evolved over millions of years to its present composition. The pie chart shows the composition of the atmosphere at one stage of its evolution.



- (i) Describe the **main** differences in the composition of the atmosphere shown in the pie chart and that present on the Earth today. [3]

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.....

- (ii) **Complete the table** to describe simple chemical tests that can be used in the laboratory to identify hydrogen and carbon dioxide. [2]

Gas	Test carried out	Expected observation
hydrogen		
carbon dioxide		



- (b) During the last 250 years the level of carbon dioxide in the atmosphere has increased. Most scientists believe that the increase in the level of carbon dioxide in the atmosphere has resulted in global warming.

The table below shows some data about the atmosphere between 1750 and 2000.

Year	Concentration of carbon dioxide in the atmosphere (parts per million)	Average global temperature (°C)
1750	278	13.3
1800	282	13.4
1850	288	13.5
1900	297	13.7
1950	310	14.0
2000	368	14.6

Use the data in the table to answer parts (i) and (ii).

- (i) Describe the trend in average global temperature between 1750 and 2000. [2]

.....

.....

.....

- (ii) Calculate the percentage increase in the concentration of carbon dioxide in the atmosphere from 1750 to 1850. [2]

percentage increase = %



8. (a) The following table gives information about some atoms, **A-E**.

A-E are not the chemical symbols of the elements.

Atom	Number of protons	Number of neutrons
A	12	12
B	10	10
C	6	8
D	11	12
E	6	6

- (i) Give the letter, **A-E**, of the atom with a mass number of 12. [1]

.....

- (ii) Give the letter, **A-E**, of the atom found in Group 1 of the Periodic Table. [1]

.....

- (iii) Choose the letters, **A-E**, of the atoms that are isotopes. Give the reason for your choice. [2]

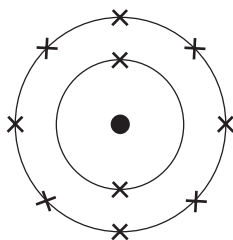
Letters and

Reason

.....



(b) The diagram below shows the electronic structure of an element in the Periodic Table.



(i) Using × to represent an electron, draw a diagram to show the electronic structure of the element which lies directly **below** this one in the Periodic Table. [1]

(ii) Explain why both of these elements are chemically unreactive. [1]

.....

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



24

Group



4



4

O

1	H	1	Hydrogen
---	---	---	----------

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88

45	Sc	Scandium	21	48	Ti	Titanium	22	51	V	Vanadium	23	52	Cr	Chromium	24	55	Mn	Manganese	25	56	Fe	Iron	26	59	Co	Cobalt	27	59	Ni	Nickel	28	63.5	Cu	Copper	29	65	Zn	Zinc	30
89	Y	Yttrium	39	91	Zr	Zirconium	40	93	Nb	Niobium	41	96	Mo	Molybdenum	42	99	Tc	Technetium	43	101	Ru	Ruthenium	44	103	Rh	Rhodium	45	106	Pd	Palladium	46	108	Ag	Silver	47	112	Cd	Cadmium	48
139	La	Lanthanum	57	179	Hf	Hafnium	72	181	Ta	Tantalum	73	184	W	Tungsten	74	186	Re	Rhenium	75	190	Os	Osmium	76	192	Ir	Iridium	77	195	Pt	Platinum	78	197	Au	Gold	79	201	Hg	Mercury	80

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

Key

A_r	Symbol	Name	Z
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relative atomic mass

— atomic number

Surname	Centre Number	Candidate Number
Other Names		0



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3430U20-1



SCIENCE (Double Award)

Unit 2: CHEMISTRY 1
FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	9	
3.	7	
4.	6	
5.	10	
6.	6	
7.	8	
8.	7	
Total	60	

ADDITIONAL MATERIALS

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INFORMATION FOR CANDIDATES

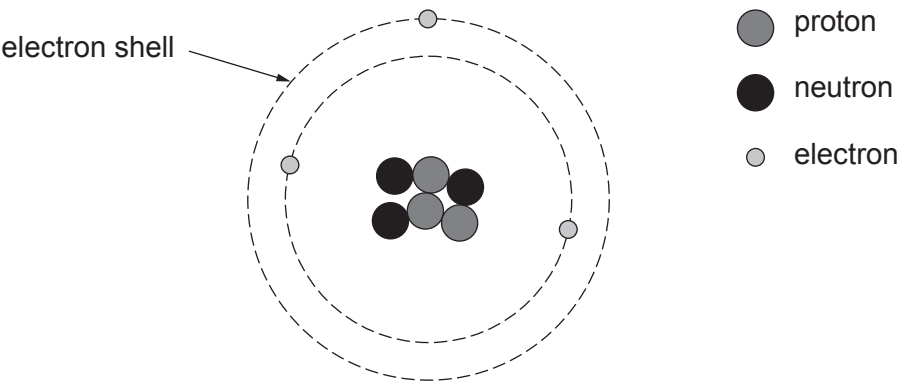
The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



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Answer all questions.

1. (a) The diagram represents a lithium atom. Its nucleus contains protons and neutrons.



(i) Complete the table of information about particles in the atom. [2]

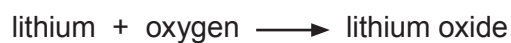
Name of particle	Charge	Mass
proton		1
neutron	0	
electron	-1	almost zero

(ii) Use the information in the diagram to give the mass number of this lithium atom. [1]

.....



(b) Lithium reacts with oxygen. The word equation for this reaction is as follows.



(i) Name the **metal** in this equation. [1]

.....

(ii) Name the **compound** in this equation. [1]

.....

(iii) Name a **reactant** in this equation. [1]

.....

(iv) Lithium oxide contains the ions Li^+ and O^{2-} .

Underline the formula of lithium oxide. [1]



2. Peter, Claire and David compared the hardness of three samples of water, **A**, **B** and **C**. They added soap solution to each sample until it gave a lather on shaking. They carried out a fair test.

(a) Give **two** ways they made it a fair test. [2]

.....

.....

.....

(b) Their results are shown in the table.

Water sample	Volume of soap solution needed to give a lather (cm ³)			
	Peter	Claire	David	Mean
A	18	19	17	
B	8	10	9	9
C	1	1	1	1

- (i) Calculate the mean volume of soap solution needed for sample **A** to give a lather. **Show your working.** [2]

Mean = cm³

- (ii) Which of the three water samples is the **hardest**? Give a reason for your answer. [2]

.....

.....

.....



(c) The box shows some advantages and some disadvantages of hard water.

strengthens bones and teeth	forms limescale when heated
wastes soap	reduces heart disease
forms a scum with soap	tastes better

Choose **all** the disadvantages of hard water from the box.

[2]

.....

.....

.....

.....

(d) Choose from the box an ion that causes hardness in water.

[1]

K⁺	Na⁺	Ca²⁺	Cu²⁺
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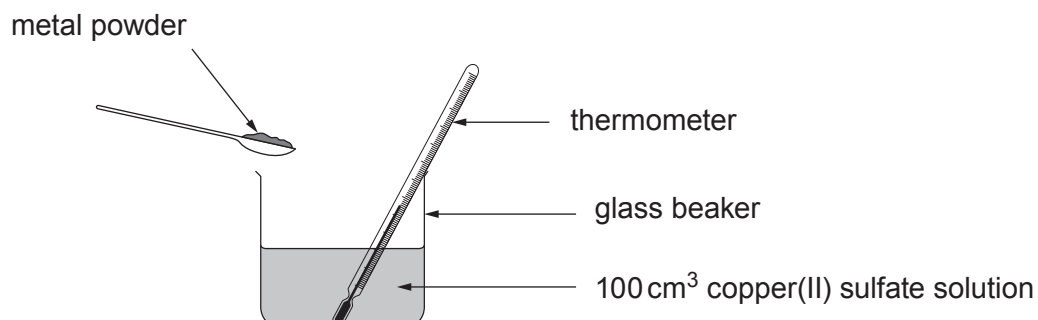
Ion

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05

9



3. Magnesium, zinc and iron powders were each added separately to 100 cm³ of copper(II) sulfate solution, to see which gave the greatest temperature change.



The temperature was recorded before and after each reaction. The results are shown in the table.

Metal	Temperature before the reaction (°C)	Temperature after the reaction (°C)	Temperature increase (°C)
zinc	20		14
magnesium	19	39	20
iron	19	24	5

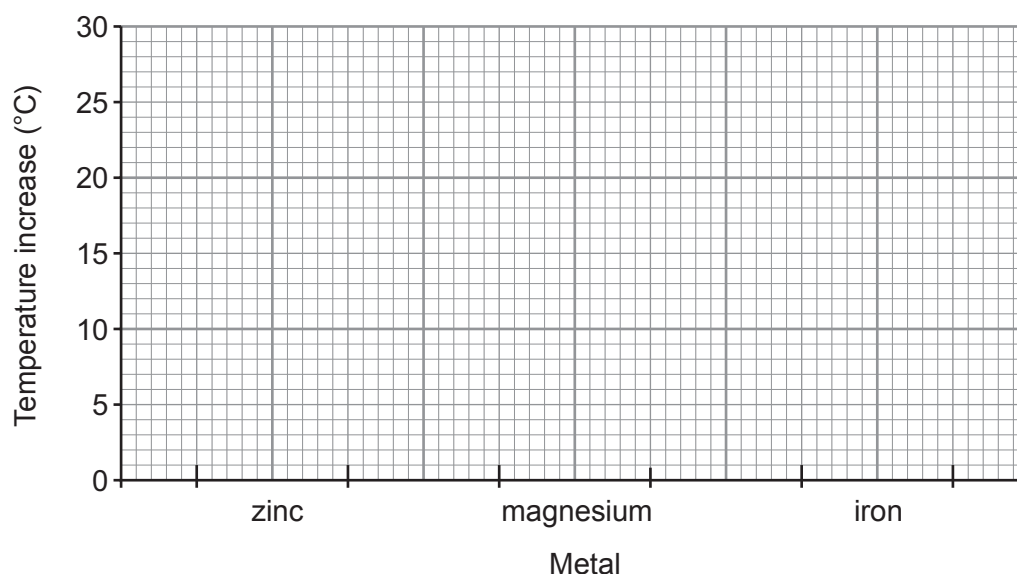
- (a) Calculate the temperature after the reaction with zinc. [1]

Temperature = °C



- (b) Plot a **bar chart** to show the temperature increase for each metal.

[2]



- (c) (i) Calculate the energy released in the reaction with magnesium. Use the following equation.

[2]

$$\begin{array}{ccccccc} \text{energy released} & = & \text{volume of solution} & \times & 4.2 & \times & \text{temperature increase} \\ \text{(J)} & & \text{(cm}^3\text{)} & & & & \text{(}^\circ\text{C)} \end{array}$$

Energy released = J

- (ii) In this reaction magnesium sulfate, MgSO_4 , is formed. What is the relative formula mass (M_r) of magnesium sulfate?

[2]

$$A_r(\text{Mg}) = 24 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

M_r =



4. A student investigated changes that happened when pairs of colourless solutions were mixed. The table shows the observations made after mixing.

Pair of solutions	Appearance of the reactants	Appearance when mixed	Temperature change when mixed
A	two colourless solutions	no change	increase
B	two colourless solutions	bubbles form	increase
C	two colourless solutions	white precipitate forms	no change
D	two colourless solutions	no change	no change

- (a) Which pair of solutions do **not** react? Give a reason for your answer. [2]

.....

.....

.....

- (b) (i) The gas given off when pair **B** react is carbon dioxide. Describe the test for carbon dioxide gas. [1]

.....

.....

- (ii) One of the solutions in pair **B** is sodium carbonate. Give the formula of sodium carbonate. [1]

.....

- (c) (i) Which pair of solutions is silver nitrate and sodium chloride? [1]

.....

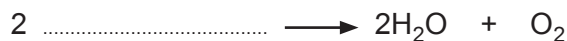
- (ii) What colour would sodium chloride give in a flame test? [1]

.....

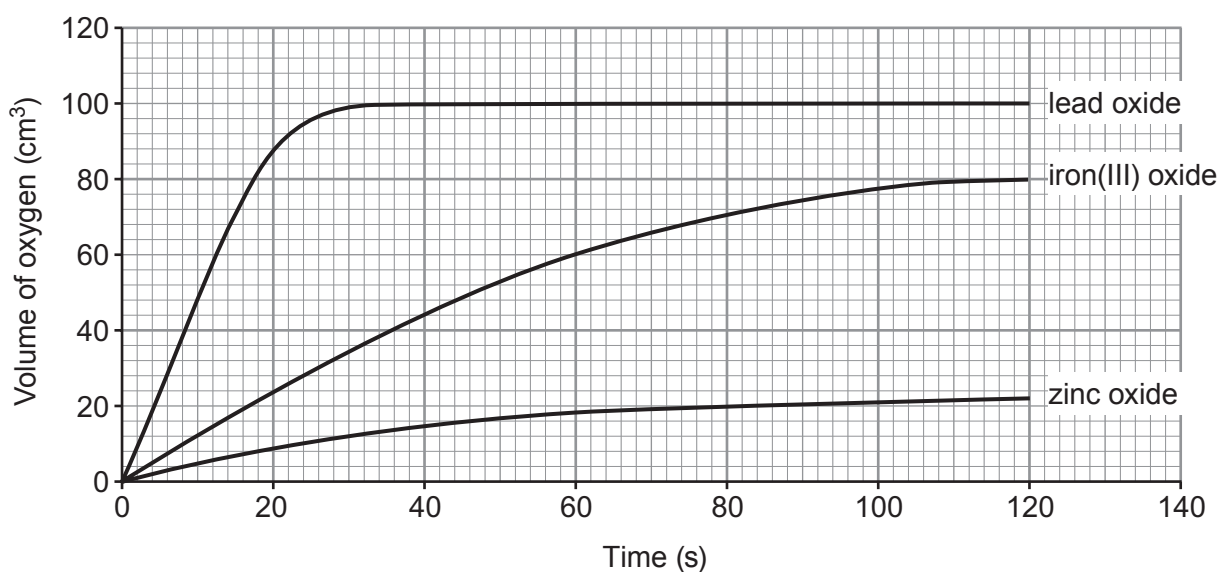


5. Hydrogen peroxide solution decomposes to give water and oxygen gas. Many metal oxides act as catalysts for the reaction.

(a) Use the symbol equation for the reaction to find the formula of hydrogen peroxide. [1]



- (b) Catalysts are used to speed up the reaction. The graphs show the volume of oxygen formed over 120 seconds when 1.0 g of different metal oxides were added to 50 cm³ of hydrogen peroxide solution.



- (i) Use the graph to find the time taken to form 44 cm³ of oxygen using iron(III) oxide as the catalyst. [1]

..... s



- (ii) State which metal oxide is the **best** catalyst. Give a reason for your answer. [1]

.....

.....

.....

- (iii) After 120 seconds, the contents of each beaker were washed into a filter paper and funnel. The catalyst left over on each filter paper was dried and weighed.

Tick (✓) the box next to the statement which is correct. [1]

the same mass of all catalysts is left over

☐

more zinc oxide is left over than lead oxide

☐

about 80 % of the iron(III) oxide is left over

☐

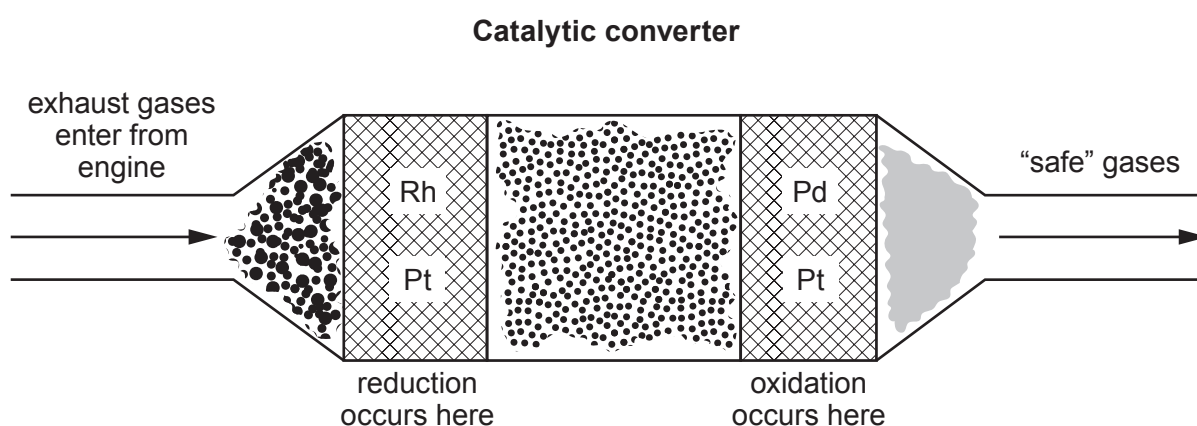
no lead oxide is left over

☐

- (c) Exhaust gases from car engines contain harmful molecules, for example, carbon monoxide and nitrogen oxides. All cars are fitted with catalytic converters that split up these harmful molecules.

The catalysts are made from platinum (Pt) and palladium (Pd) or rhodium (Rh). A mesh structure is used that exposes the maximum surface area of catalyst to the exhaust gases, while also reducing the amount of catalyst required. Platinum, palladium and rhodium are extremely expensive.

As the exhaust gases from the engine pass over the catalysts, chemical reactions take place on their surfaces. The harmful molecules are broken up and converted into other gases that are “safe” to enter the air. These gases include carbon dioxide, nitrogen, oxygen and water.



There are two different types of catalyst, a **reduction catalyst** and an **oxidation catalyst**. The reduction catalyst removes oxygen from nitrogen oxides to make nitrogen and oxygen. The oxidation catalyst adds oxygen to carbon monoxide to make carbon dioxide.

One of the biggest drawbacks of the catalytic converter is that it only works effectively at high temperatures. The average running temperature of a car engine is between 90 and 110°C. It takes nearly 30 minutes for these temperatures to be reached.

The table opposite shows the percentages of carbon monoxide and nitrogen oxides converted to safe gases by a catalytic converter at different temperatures.



Temperature (°C)	Carbon monoxide converted (%)	Nitrogen oxides converted (%)
25	16	25
50	19	28
75	26	35
100	60	72
125	91	92
150	93	94
175	95	95
200	97	98

- (i) Using your knowledge of particle theory suggest why a catalyst in mesh form works better than a lump of catalyst. [2]

.....

.....

.....

.....

- (ii) Tick (✓) the box that best describes the adverse effect that gases leaving the exhaust would have on the environment. [1]

they have no effect on the environment

☐

they reduce oxygen levels

☐

they deplete the ozone layer

☐

they cause global warming

☐

- (iii) Tick (✓) the box that best describes the conversion of carbon monoxide and nitrogen oxides into "safe" gases at different temperatures. [1]

equal amounts of carbon monoxide and nitrogen oxides
are converted at every temperature

☐

more carbon monoxide is converted than nitrogen oxides
up to 100 °C

☐

more nitrogen oxides are converted than carbon monoxide
up to 100 °C

☐

40 % more nitrogen oxides are converted than carbon monoxide
up to 100 °C

☐

- (iv) In your opinion how effective are catalytic converters? Explain your answer. [2]

.....

.....

.....

.....



Examiner
only

6. Describe how respiration and photosynthesis keep the carbon dioxide and oxygen content of the atmosphere approximately constant. Discuss how human activity is threatening this balance.

[6 QER]

6	



Examiner
only

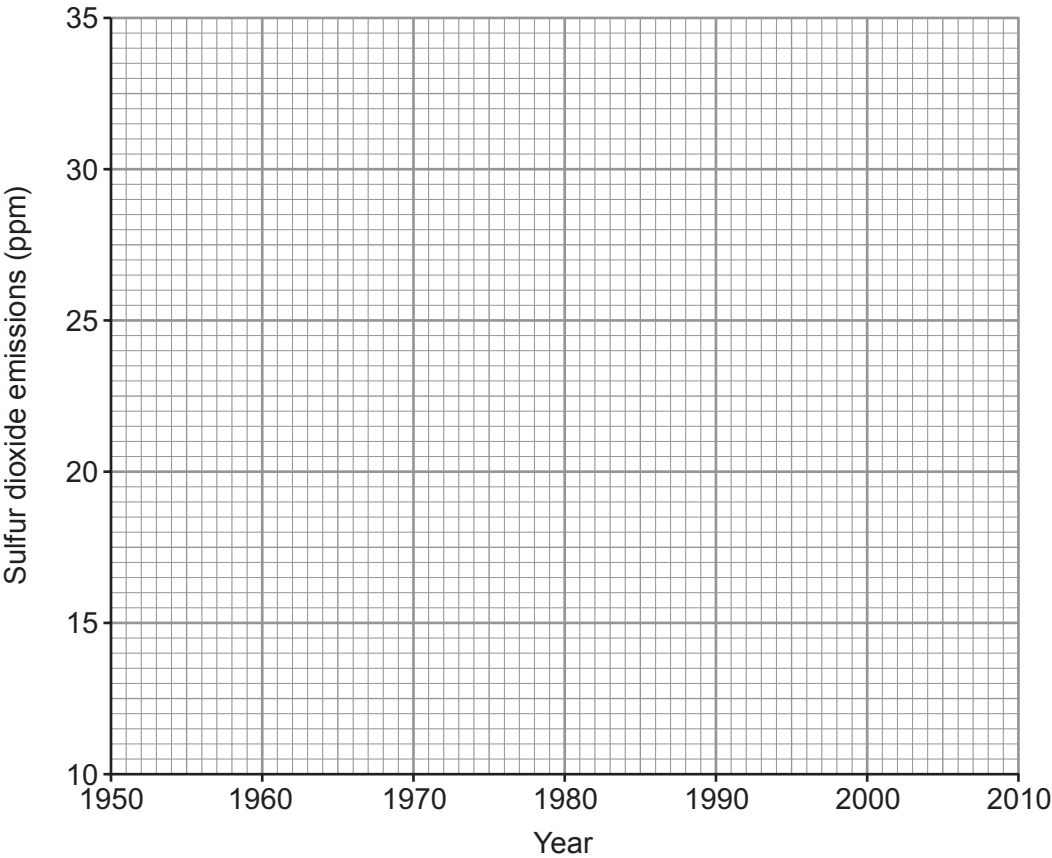
7. Burning fossil fuels containing sulfur causes sulfur dioxide, SO₂, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

(a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



- (ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

.....

.....

.....

- (iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

.....

.....

.....

- (b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H₂S.

Complete and balance the symbol equation for this reaction. [2]



8. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Electrical conductor
oxygen	−219	−183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

.....

(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

.....

.....

(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

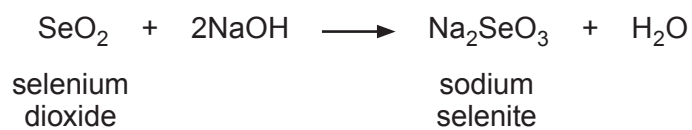
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.....



(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



(i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3 . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$M_r =$

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage = %

7

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1 H Hydrogen 1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

24

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3430U20-1



S19-3430U20-1

WEDNESDAY, 12 JUNE 2019 – MORNING**SCIENCE (Double Award)****Unit 2: CHEMISTRY 1
FOUNDATION TIER**

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen.
Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	8	
3.	6	
4.	7	
5.	6	
6.	6	
7.	7	
8.	8	
9.	7	
Total	60	

3430U201
01

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1. Elements **A**, **B**, **C**, **D** and **E** are shown in the Periodic Table.

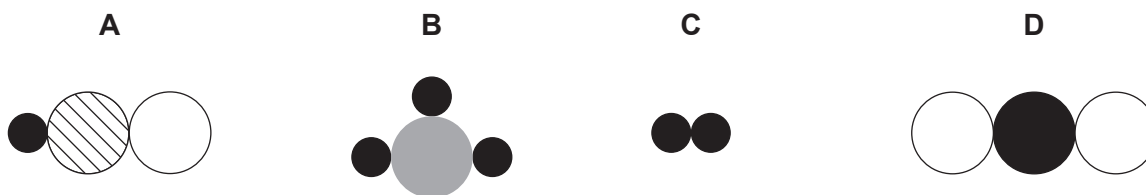
[illegible]

[5]

	Letter
is in Group 2	
makes a 'pop' noise with a lit splint	
has both metal and non-metal properties	
reacts with lithium to make lithium chloride	
is unreactive	



2. Diagrams **A**, **B**, **C** and **D** represent hydrogen (H_2), sulfur dioxide (SO_2), hypochlorous acid (HClO) and phosphine (PH_3) but **not in that order**.



- (a) (i) Give the **letter** of the diagram that represents an element. Give a reason for your answer. [2]

Letter

Reason

- (ii) Give the **letter** of the diagram that represents SO_2 . [1]

.....

- (iii) Use diagrams **A**, **B**, **C** and **D** above to work out which of the following diagrams represents water (H_2O). **Circle** the correct diagram. [1]



- (b) (i) Calculate the relative molecular mass (M_r) of hypochlorous acid, HClO. [1]

$$A_r(\text{H}) = 1$$

$$A_r(\text{O}) = 16$$

$$A_r(\text{Cl}) = 35.5$$

$$M_r = \dots\dots\dots$$

- (ii) The relative molecular mass (M_r) of sulfur dioxide is 64.

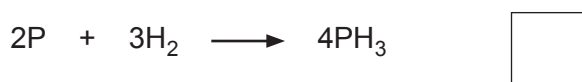
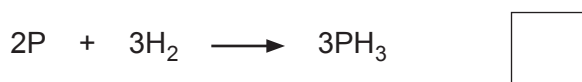
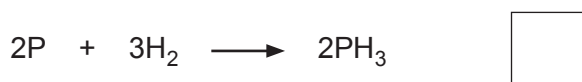
Calculate the percentage by mass of sulfur in sulfur dioxide, SO₂. [2]

$$A_r(\text{S}) = 32$$

$$\text{Percentage} = \dots\dots\dots \%$$

- (c) Phosphine (PH₃) is made when phosphorus and hydrogen react.

Put a tick (✓) in the box next to the correctly balanced symbol equation for this reaction. [1]

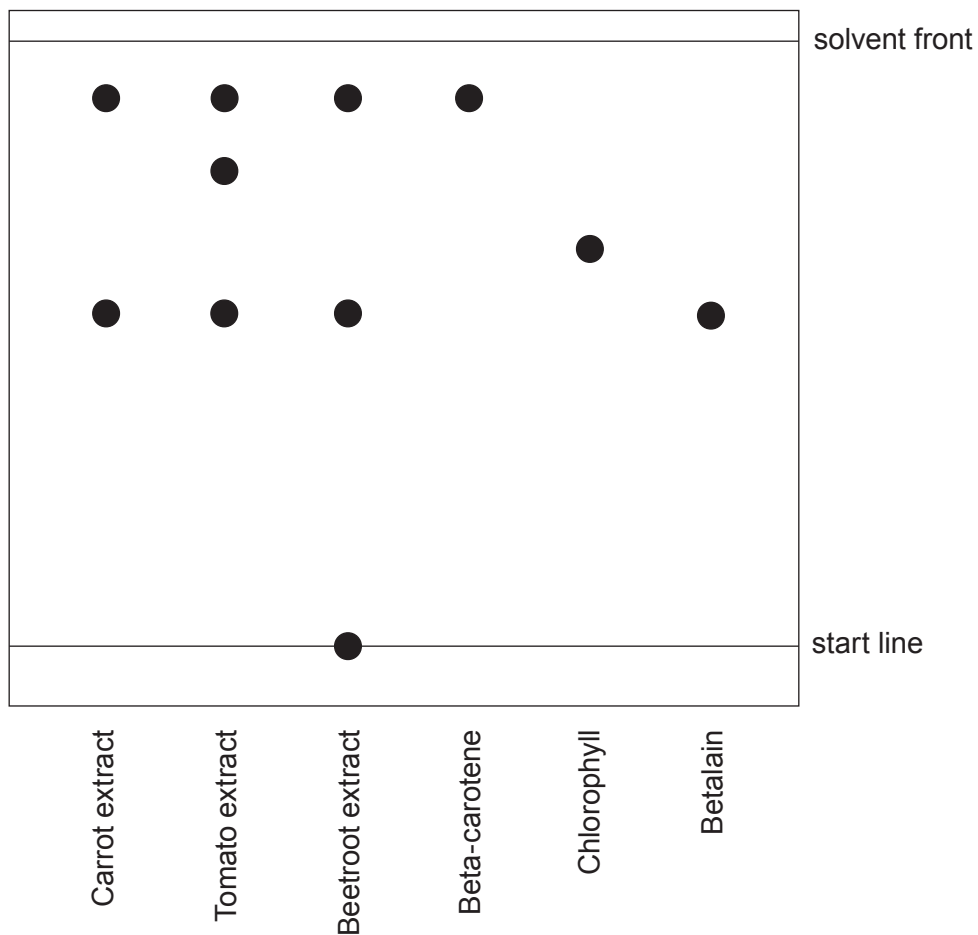


3. A student investigates the pigments found in some fruit and vegetables.

She obtains some coloured extracts from carrots, tomatoes and beetroot.

She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and betalain.

The diagram shows the chromatogram at the end of the experiment.



- (a) Give the names of **all** the substances which are **not** mixtures. Give a reason for your choice. [1]

Substances

Reason



- (b) Put a tick (✓) in the boxes next to the **two** conclusions that can be drawn from the chromatogram. [2]

chlorophyll is not present in carrot, tomato or beetroot extracts

☐

beta-carotene is present in carrot extract but not present in tomato extract

☐

both beta-carotene and betalain are present in beetroot extract

☐

betalain is present in tomato extract but not present in carrot extract

☐

both carrot and beetroot extracts contain a pigment other than beta-carotene, chlorophyll and betalain

☐

- (c) One of the pigments present in the carrot extract has travelled 4.4 cm above the start line.

The solvent front has travelled 8.0 cm. Calculate the R_f value of the pigment using the following equation. [2]

$$R_f = \frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent front}}$$

$R_f =$

- (d) Give the reason why there is a spot remaining on the start line in the chromatogram for beetroot. [1]

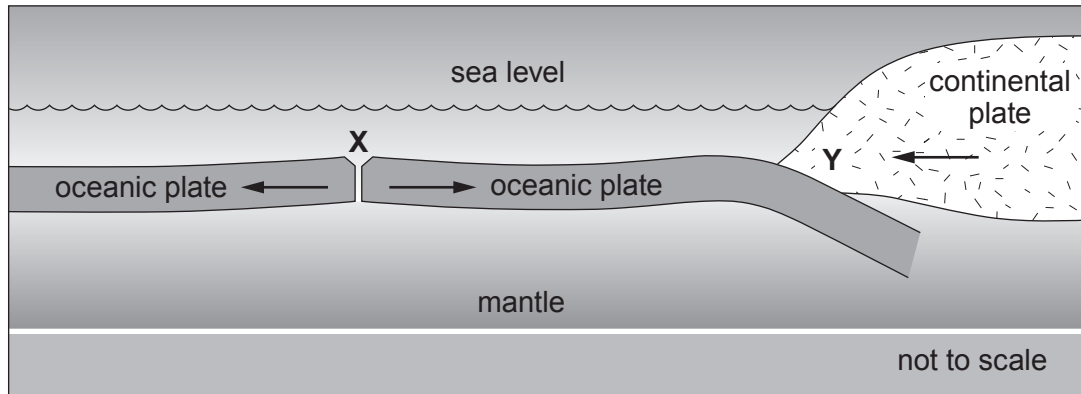
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4. The Earth's crust is broken up into tectonic plates.

The diagram shows two plate boundaries **X** and **Y**.



- (a) Describe what happens as the plates move apart at plate boundary **X**.

[2]

.....

.....

.....



- (b) The two types of plate have different densities as shown in the table.

Type of plate	Density (g/cm ³)
continental	4.7
oceanic	5.0

At boundary **Y** the oceanic plate is pushed **underneath** the continental plate.

- (i) State why the oceanic plate is pushed underneath the continental plate. [1]

.....

- (ii) State what happens to the oceanic plate when it goes underneath the continental plate. [1]

.....

- (iii) Underline the name of the type of plate boundary seen at **Y**. [1]

destructive

constructive

conservative

- (c) Using satellite imaging, scientists discovered that one oceanic plate had moved 537 cm over 600 years. Calculate the mean rate of movement of the plate over this time. [2]

$$\text{rate} = \frac{\text{distance}}{\text{time}}$$

Rate = cm/year

3430U201
09



5. Planet **J** is similar in size to the Earth. However, the temperature on planet **J** is about 470 °C and the clouds in its atmosphere are made of sulfuric acid.

A group of students investigated the properties of some metals. Their aim was to see if they could find a metal suitable for designing a spacecraft to explore planet **J**. Their findings are shown below.

Zinc – fizzes quite vigorously with sulfuric acid, has a melting point of 420 °C and a density of 7.1 g/cm³

Copper – has a melting point of 1083 °C, does not react with sulfuric acid and has a density of 8.9 g/cm³

Sodium – has a density of 1.0 g/cm³, reacts explosively with sulfuric acid and has a melting point of 98 °C

Magnesium – has a density of 1.7 g/cm³, a melting point of 650 °C and it fizzes vigorously with sulfuric acid

Titanium – does not react with sulfuric acid, has a melting point of 1675 °C and a density of 4.5 g/cm³

Lead – has a melting point of 328 °C, a density of 11.3 g/cm³ and does not react with sulfuric acid

The spacecraft needs to withstand the conditions on the surface of planet **J**. The mass of the spacecraft also needs to be as low as possible in order for it to have enough energy to escape the Earth's gravity.

- (a) Which **one** of these statements best describes why magnesium is an **unsuitable** metal for the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

its density is 1.7 g/cm³

☐

its melting point is 650 °C

☐

it fizzes vigorously with sulfuric acid

☐

it is malleable

☐

- (b) Small amounts of lead are sometimes used in electrical circuits.

Which **one** of these statements best describes why lead would **not** be suitable for use in the electrical circuits of the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

it does not react with sulfuric acid

☐

it is ductile

☐

it would melt when it lands on planet J

☐

its density is 11.3g/cm³

☐

- (c) The students decided that titanium is the most suitable metal from which to build the spacecraft.

Put a tick (✓) in the boxes next to the **two** statements that best describe the reasons for their choice. [2]

it does not react with sulfuric acid

☐

it is expensive

☐

it is a good conductor of heat

☐

it is non-magnetic

☐

it has a melting point much higher than the temperature on planet J

☐

it is shiny so will reflect the sun's rays

☐

- (d) Sodium reacts explosively with sulfuric acid. Sodium sulfate and hydrogen are produced. Complete and balance the equation for this reaction. [2]



Examiner
only

6. Several areas of the UK add fluoride to drinking water.

State the benefit of fluoridation and discuss the reasons why some people are opposed to it.

[6 QER]

.....

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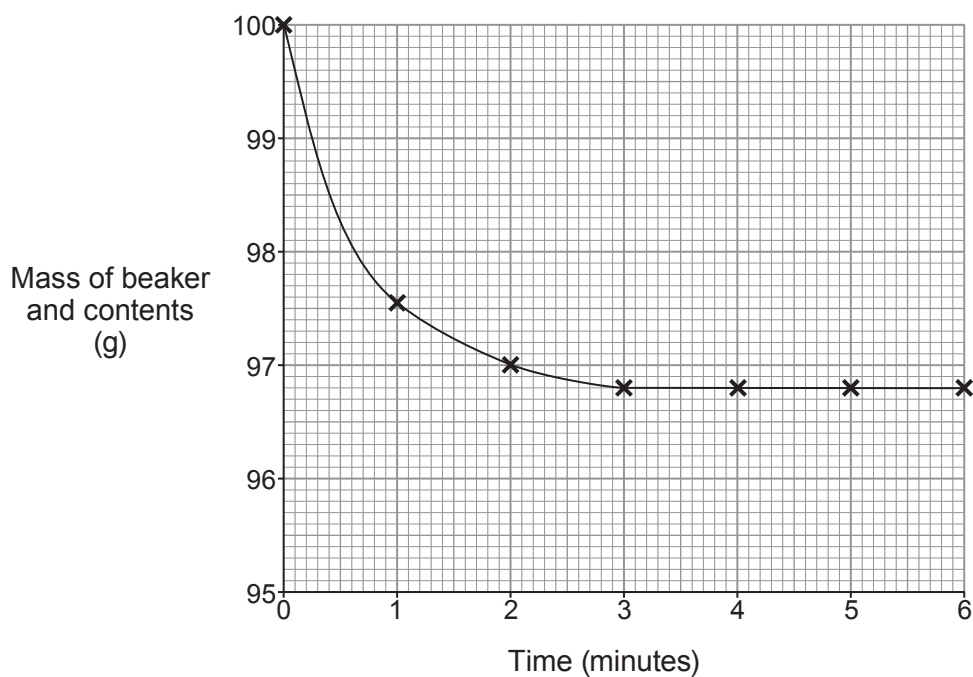
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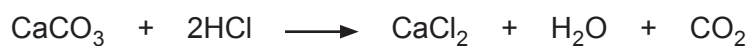
6



7. Powdered calcium carbonate was added to an excess of dilute hydrochloric acid in a beaker. The mass of the beaker and its contents was recorded every minute for 6 minutes. The graph shows the results.



The equation for the reaction is



- (a) Describe how the mass of the beaker and its contents changes over the first minute. Give the reason for this change. [2]

.....

.....



- (b) (i) Use the graph to determine the time taken for the reaction to finish. [1]

- (ii) When is the reaction at its fastest? Tick (✓) the correct box. [1]

from 0 – 0.5 minutes

☐

from 1 – 1.5 minutes

☐

from 2 – 2.5 minutes

☐

from 3 – 3.5 minutes

☐

- (c) Use your graph to calculate the mean rate of the reaction during the **first two minutes**.
Use the following equation. [2]

$$\text{rate} = \frac{\text{decrease in mass of beaker and contents (g)}}{\text{time (minutes)}}$$

Rate = g/minute

- (d) The experiment was repeated using the same mass of calcium carbonate but as a **lump** instead of a powder.

On the grid opposite, sketch the graph you would expect to obtain from this second experiment. [1]

7



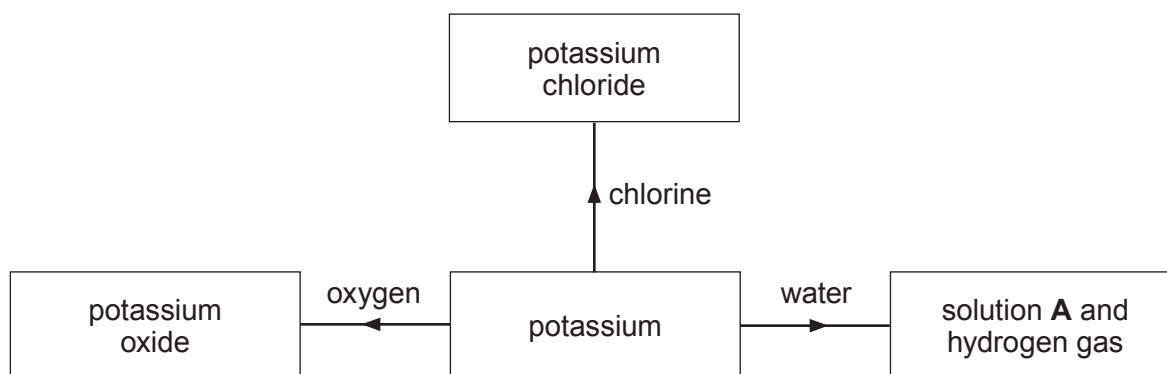
8. (a) The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine		7	3
.....	2,8,5	5	3
potassium	2,8,8,1	1	

Complete the table.

[3]

- (b) The flow chart shows some of the reactions of potassium.



- (i) State **one** observation you would make when potassium reacts with water. [1]

.....

.....

- (ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

.....

.....



(iii) Give the formula of solution **A**.

[1]

.....

(iv) Suggest a value for the pH of solution **A**.

[1]

.....

(v) Name a Group 1 metal that is **more** reactive than potassium.

[1]

.....

Examiner
only

8



9. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

(i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

.....

.....

.....

(ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

.....

.....

.....

(b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

.....

.....

.....

.....

.....

.....

(c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1
H
Hydrogen
1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

24

65 Zn Zinc 30	63.5 Cu Copper 29	59 Ni Nickel 28	59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	52 Cr Chromium 24	51 V Vanadium 23	48 Ti Titanium 22	45 Sc Scandium 21
112 Cd Cadmium 48	108 Ag Silver 47	106 Pd Palladium 46	103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	96 Mo Molybdenum 42	93 Nb Niobium 41	91 Zr Zirconium 40	89 Y Yttrium 39
201 Hg Mercury 80	197 Au Gold 79	195 Pt Platinum 78	192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	184 W Tungsten 74	181 Ta Tantalum 73	179 Hf Hafnium 72	139 La Lanthanum 57

Key

Ar
Symbol
Name
Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3430U20-1



Z22-3430U20-1

FRIDAY, 17 JUNE 2022 – AFTERNOON**SCIENCE (Double Award)****Unit 2 – CHEMISTRY 1
FOUNDATION TIER**

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **7** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	11	
3.	6	
4.	5	
5.	6	
6.	5	
7.	6	
8.	9	
9.	6	
Total	60	

3430U201
01

JUN223430U20101

Answer **all** questions.

1. (a) Rock salt is a mixture of salt and sand. Crystals of pure salt can be obtained from rock salt.

A–E are the steps in the method used but they are in the wrong order.

- A** Add water to a sample of rock salt in a beaker and stir
- B** Heat the solution to evaporate some of the water
- C** Grind the rock salt into a fine powder
- D** Filter the mixture to separate the sand from the salt solution
- E** Leave the saturated solution in a warm place for a few days so that crystals of salt form

Put the steps in the correct order. The first step is already included.

[2]

C

first step

- (b) A student was asked to investigate the dyes present in an orange sweet.

The student carried out the following method. There are two errors in the method.

- Draw a line using a ruler and pen on chromatography paper.
- Place a sample of the orange colour on the line.
- Stand the chromatography paper in a beaker and add enough water to just cover the sample.
- Leave the paper to stand until the water rises to the top of the paper.

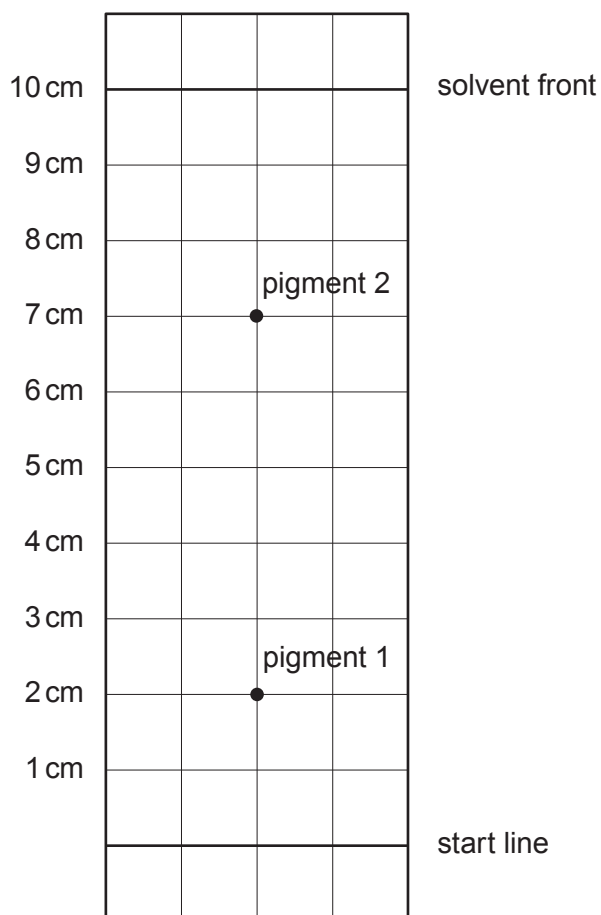
- (i) State the **two** errors in the method.

[2]

1.
2.



(ii) Another student used a correct method and obtained the chromatogram below.



Use the formula to calculate the R_f value for pigment 2.

[2]

$$R_f = \frac{\text{distance travelled by the pigment}}{\text{distance travelled by the solvent front}}$$

$R_f = \dots\dots\dots$

6

3430U201
03



2. A year 10 class investigated the reactions between some metals and hydrochloric acid. Their results are summarised in the table below.

Metal	Initial temperature (°C)	Final temperature (°C)	Rise in temperature (°C)	General observations
zinc	21	32	11	a few bubbles
calcium	22	66	44	lots of bubbles, solution spills out of test tube
magnesium	20		31	lots of bubbles
copper	21	21	0	no bubbles
iron	22	25	3	one or two bubbles

- (a) (i) State which metal did **not** react with hydrochloric acid.

Give a reason for your choice.

[1]

Metal

Reason

- (ii) Calculate the final temperature for the reaction between magnesium and hydrochloric acid.

[1]

Final temperature = °C

- (b) What name is given to a reaction which gives a rise in temperature?

[1]

Choose your answer from the box.

endothermic

combustion

exothermic

precipitation

.....



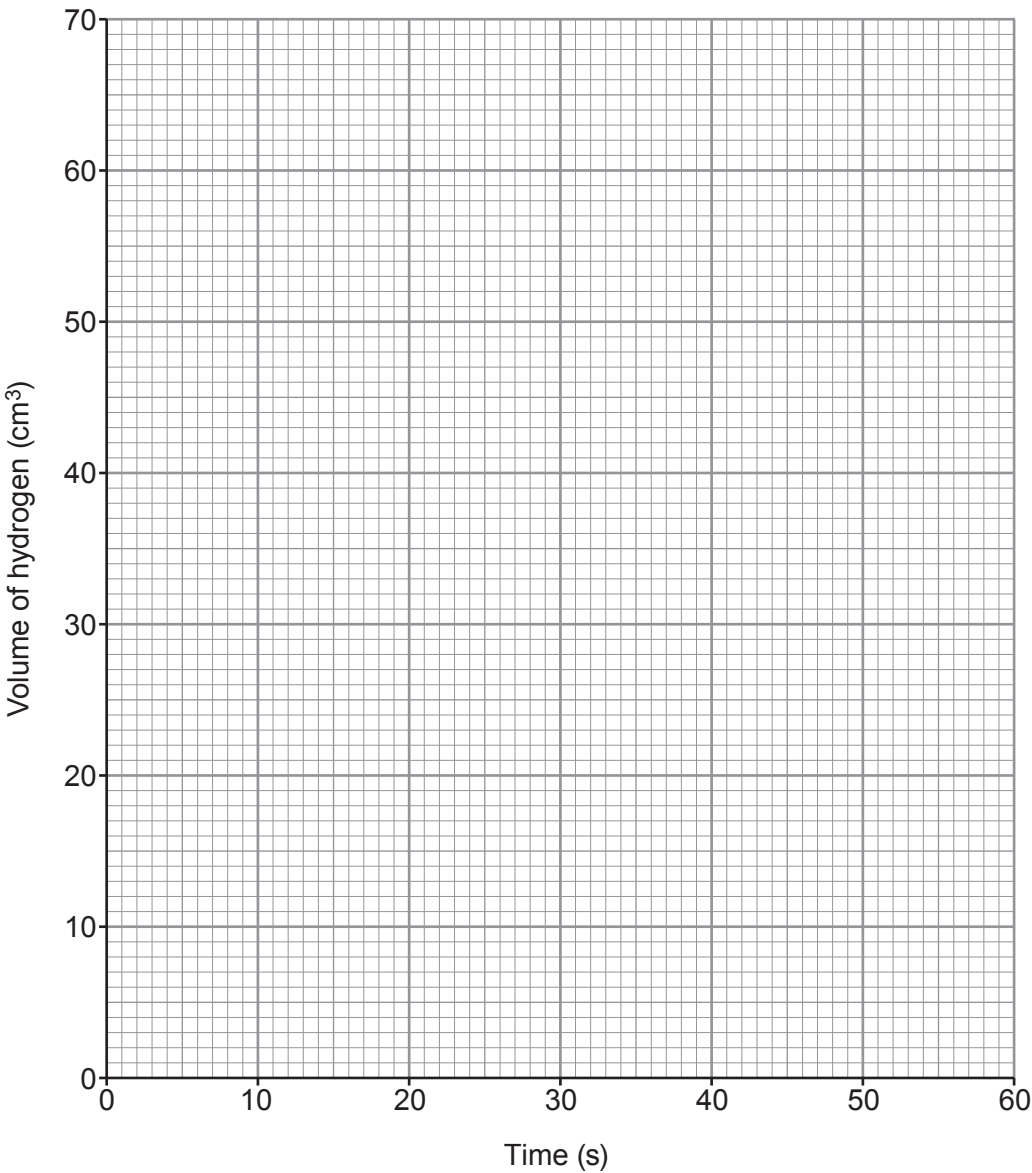
- (c) The year 10 class then decided to investigate the rate of the reaction between magnesium and hydrochloric acid.

A piece of magnesium was placed in excess hydrochloric acid at 20 °C. The volume of hydrogen produced was recorded every 10 s.

The results obtained are shown in the table.

Time (s)	0	10	20	30	40	50	60
Volume of hydrogen (cm ³)	0	19	31	40	47	53	56

- (i) Plot the volume of hydrogen against time on the grid. Draw a suitable line. [3]



- (ii) The reaction had not finished after 60 s. How does the graph show this?

Put a tick (✓) in the correct box.

[1]

Graph stops at 60 s

☐

Graph is still rising at 60 s

☐

Graph reaches a maximum temperature of 56 °C

☐

- (iii) Why does the reaction slow down over time?

Put a tick (✓) in the correct box.

[1]

The particles collide with less energy so less chance of successful collisions

☐

The particles move slower so less chance of successful collisions

☐

The particles have less surface area so less chance of successful collisions

☐

The particles get used up so less chance of successful collisions

☐

- (iv) Suggest **two** changes you could make to the hydrochloric acid to make the reaction faster.

[2]

.....

.....

- (v) In the reaction between magnesium and hydrochloric acid, magnesium chloride is formed. Magnesium chloride contains Mg^{2+} ions and Cl^- ions.

Give the formula of magnesium chloride.

[1]

.....



3. A beryllium atom can be represented by the following symbol.



- (a) Choose a number from the box to complete each of the following sentences. [4]

4	5	9	13
---	---	---	----

Each number can be used once, more than once or not at all.

Beryllium has protons.

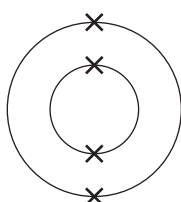
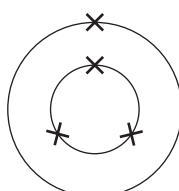
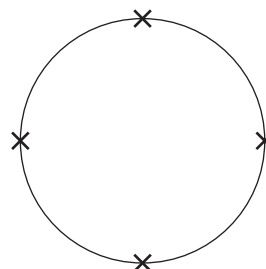
The atomic number of beryllium is

The mass number of beryllium is

Beryllium has neutrons.

- (b) Beryllium has 4 electrons.

Put a tick (✓) in the box that shows the correct electronic structure for beryllium. [1]


☐

☐

☐

- (c) Beryllium is found in the compound beryllium fluoride, BeF_2 .

Give the total number of atoms shown in the formula. [1]

.....



4. The table shows various stages in the development of the Earth's atmosphere since its formation 4 500 million years ago.

Stage	Major events	Gases present in the atmosphere
1	volcanic eruptions	carbon dioxide, water vapour, methane, ammonia
2	oceans form	carbon dioxide, methane, ammonia
3	green plants evolve	carbon dioxide, nitrogen, oxygen
4	most carbon dioxide becomes locked in rock and fossil fuels	nitrogen, oxygen, water vapour

- (a) Which **one** of these statements best describes how the oceans were formed?

Put a tick (✓) in the box next to the correct answer.

[1]

Water vapour evaporated to form clouds

☐

The Earth cooled so water vapour condensed

☐

Bacteria and algae turned the water vapour into liquid water

☐

There were no more volcanoes to produce water vapour

☐

- (b) Explain why the appearance of green plants was an important stage in the development of the atmosphere. [2]

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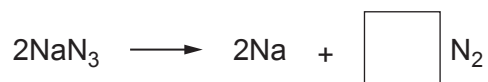
- (c) The atmosphere today contains nitrogen, oxygen and water vapour. [1]

Use the following information to identify another gas present.

- a Group 0 gas
 - the third most abundant in the atmosphere
 - used in light bulbs and as an inert atmosphere for welding
-

- (d) Nitrogen can be obtained by heating sodium azide, NaN_3 . Sodium is also produced in the reaction.

Complete the balancing of the equation for this reaction. [1]



5. Drinking water contains a number of ions which are important in the human body. Some of these ions cause hardness in water.

The table shows the concentration of ions in drinking water from four different locations.

Location	Concentration of ions (mg / dm ³ of water)					
	potassium K ⁺	ammonium NH ₄ ⁺	calcium Ca ²⁺	fluoride F ⁻	sulfate SO ₄ ²⁻	nitrate NO ₃ ⁻
A	0.1	0.4	0.0	0.0	0.4	0.2
B	0.0	0.3	0.4	4.4	0.2	0.0
C	0.2	0.6	2.7	0.4	0.0	0.1
D	3.4	2.1	1.0	2.1	2.5	2.3

- (a) Which location is likely to have the hardest water?

Put a tick (✓) in the box next to the correct answer.

[1]

A ☐

B ☐

C ☐

D ☐

- (b) In which location do people have the least protection against tooth decay from their drinking water?

Put a tick (✓) in the box next to the correct answer.

[1]

A ☐

B ☐

C ☐

D ☐



- (c) Ionic compounds dissolve in water to form positive and negative ions.

Which **two** compounds may have dissolved in the drinking water at location **B**?

Put a tick (✓) in the box next to the correct answer.

[1]

potassium fluoride and calcium sulfate

☐

ammonium sulfate and potassium nitrate

☐

calcium fluoride and ammonium nitrate

☐

ammonium sulfate and calcium fluoride

☐

- (d) (i) Calculate the relative formula mass (M_r) of calcium sulfate, CaSO_4 .

[1]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{S}) = 32$$

$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (ii) Calculate the percentage by mass of fluorine in calcium fluoride, CaF_2 .

[2]

$$M_r(\text{CaF}_2) = 78$$

$$A_r(\text{F}) = 19$$

$$\text{Percentage} = \dots\dots\dots \%$$

6



6. (a) A technician carried out a flame test and a silver nitrate test on a solution of sodium iodide.

Draw **one** line from **each** test to the correct observation.

[2]

Test	Observation
flame test	red flame
	yellow flame
	lilac flame
silver nitrate test	white precipitate
	cream precipitate
	yellow precipitate

- (b) (i) When silver nitrate and sodium iodide react, sodium nitrate and a precipitate of silver iodide are formed.

Write the formulae of sodium nitrate and silver iodide to complete the equation. [2]



- (ii) Suggest a method you could use to remove the precipitate from the reaction mixture.

[1]

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Examiner
only

7. Describe the advantages and disadvantages of hard water. [6 QER]

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8. (a) (i) State why sodium is stored in oil in the laboratory. [1]

.....

(ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

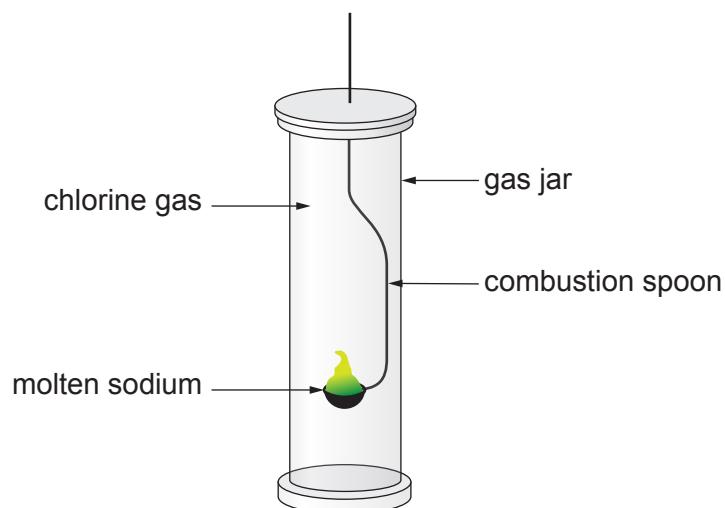
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(iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

.....

(b) The diagram shows the reaction of sodium with chlorine.



(i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

.....

(ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]



(c) The table shows some properties of Group 7 elements.

Element	Melting point (°C)	Boiling point (°C)	Reaction with hot iron
fluorine	−220	−188	explosive
chlorine	−101	−34	very fast
bromine	−7	59	quite fast
iodine	114		slow

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

−25 °C ☐ 25 °C ☐ 100 °C ☐ 150 °C ☐

(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer. [2]

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9



9. (a) In 1915 Alfred Wegener suggested that the Earth's continents were once joined together as one large land mass.



- (i) State **three** pieces of evidence that Alfred Wegener used to support his theory. [3]

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- (ii) Explain why Wegener's theory was not originally accepted by other scientists, but it is today. [2]

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.....

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.....



- (b) What type of destructive event is likely to happen at a **conservative** plate boundary? [1]

.....

Examiner
only

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

1 H Hydrogen 1																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	4 He Helium 2	
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
										Key									

Key

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE****3430U20-1**

S23-3430U20-1

FRIDAY, 16 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 2 – CHEMISTRY 1
FOUNDATION TIER

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	9	
3.	9	
4.	8	
5.	6	
6.	5	
7.	8	
8.	7	
Total	60	

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Answer **all** questions.

1. (a) The table shows four compounds and their formulae.

Compound	Formula
nitrogen dioxide	NO ₂
hydrogen bromide	HBr
ethane	C ₂ H ₆
ammonia	NH ₃

Use the information in the table to answer the following questions.

- (i) Give the **name** of the compound that contains the elements hydrogen and bromine. [1]

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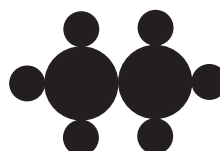
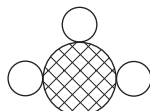
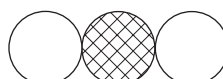
- (ii) Give the **names** of the elements present in ethane. [1]

..... and

- (iii) Give the **total** number of atoms found in a molecule of nitrogen dioxide. [1]

.....

- (iv) Tick (✓) the box next to the diagram that represents a molecule of ammonia, NH₃. [1]


☐

☐

☐

☐


- (b) Some ants spray a toxic mixture containing formic acid to defend themselves from predators.

Each molecule of formic acid contains two atoms of hydrogen, one atom of carbon and two atoms of oxygen.

Use this information to give the formula of formic acid.

[1]

.....

- (c) The box contains the names of three compounds that are made from positive and negative ions.

lithium chloride

magnesium bromide

calcium oxide

- (i) Give the formulae of the **ions** present in lithium chloride.

[1]

..... and

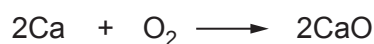
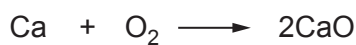
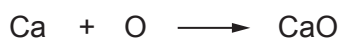
- (ii) Give the formula of magnesium bromide.

[1]

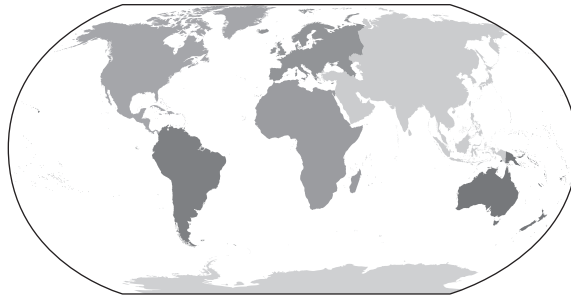
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- (iii) Tick (✓) the box next to the correct equation for the reaction between calcium and oxygen to produce calcium oxide.

[1]

☐☐☐

2. The diagram shows how the Earth's continents are arranged today.



In 1912, Alfred Wegener proposed the theory of continental drift.

(a) (i) Use statements from the table to answer parts I. and II. below.

A	there are similar animals on different continents
B	there are similar fossils on different continents
C	only Europe, Asia and Africa were once joined
D	all of the continents were once joined
E	there is a jigsaw fit of different continents
F	there are similar rocks on different continents
G	none of the continents were joined

I. Give the **letter** of the statement that describes how Wegener suggested the continents were originally arranged. [1]

.....

II. Give the **letters of three** pieces of evidence that Wegener used to support his theory. [3]

.....

(ii) Underline the correct word in the brackets to complete the following sentence. [1]

Wegener could not explain how the continents moved but it is now known that this happens because of (**conduction** / **convection** / **radiation**) currents in the Earth's mantle.



(b) The table shows how the mean temperature of the Earth’s atmosphere has increased between 1960 and 2010.

Year	Mean temperature (°C)
1960	13.9
1970	14.0
1980	14.2
1990	14.3
2000	14.4
2010	14.6

(i) Underline the mean temperature of the Earth’s atmosphere in 2020 if this trend continued. [1]

15.0 14.4 14.8

(ii) Name the gas which is mainly responsible for the temperature increase. [1]

.....

(iii) Give **two** effects of global warming. [2]

.....
.....
.....



3. The table shows how the solubility of potassium nitrate in water varies with temperature.

Temperature (°C)	Solubility (g per 100g of water)
0	13
10	21
20	32
30	46
40	64
50	86

(a) Tick (✓) the box that shows approximately how many times more soluble potassium nitrate is at 40 °C than it is at 10 °C. [1]

2 times

☐

3 times

☐

4 times

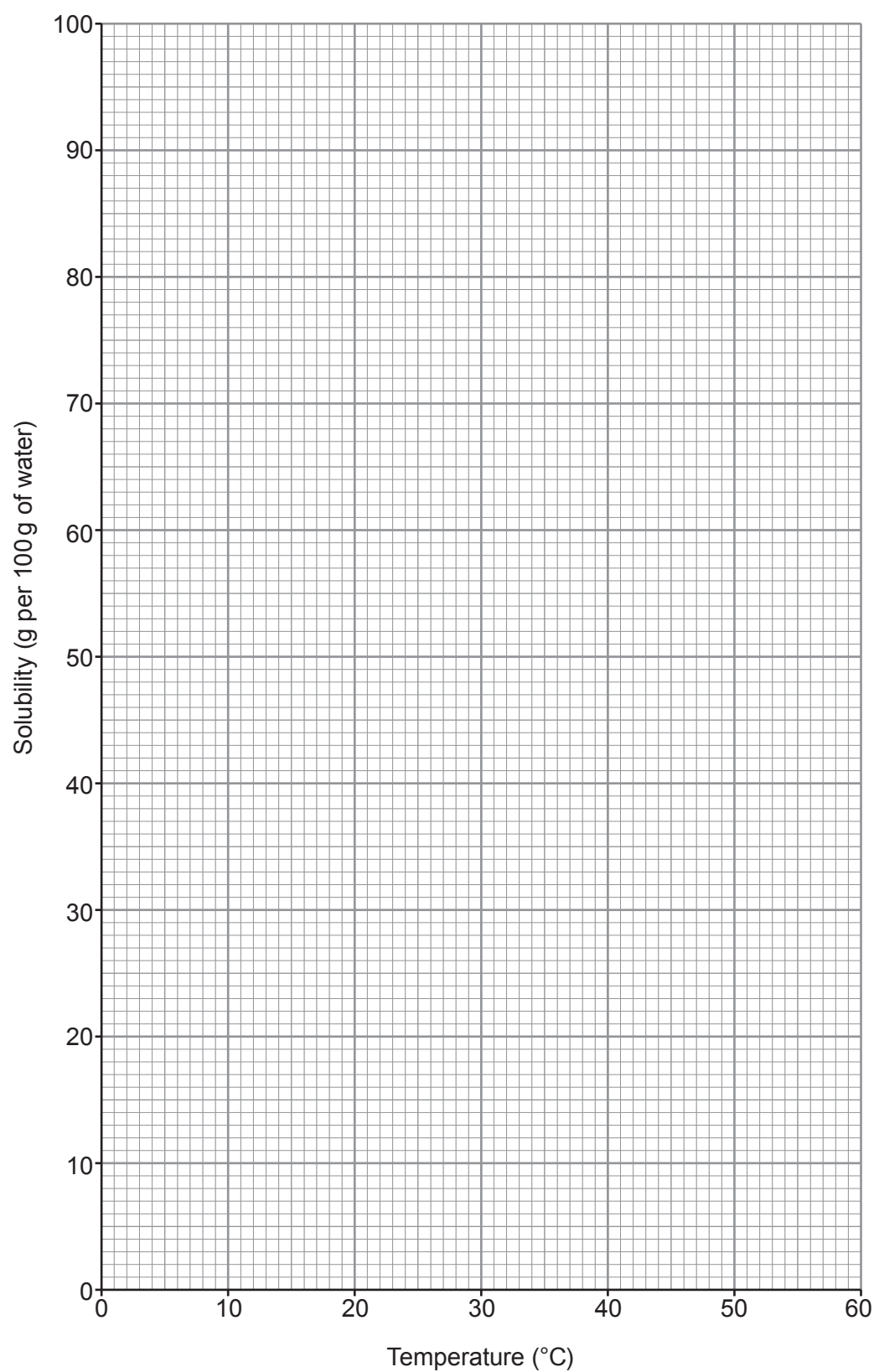
☐

5 times

☐


- (b) (i) Plot the solubility of potassium nitrate against temperature on the grid.
Draw a suitable line.

[3]



- (ii) I. State the solubility of potassium nitrate at 45 °C. [1]

..... g per 100 g of water

- II. Calculate the mass of solid potassium nitrate that forms when a saturated solution in 100 g of water at 45 °C cools to 25 °C. [1]

Mass = g

- (c) Potassium nitrate decomposes on heating to produce potassium nitrite, KNO_2 , and oxygen.

- (i) Balance the equation for the reaction. [1]



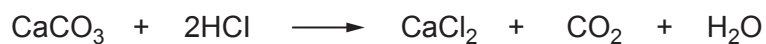
- (ii) On heating potassium nitrate, a student expected to make 4.35 g of potassium nitrite. However, in an experiment she only made 3.13 g.

Calculate the percentage yield of potassium nitrite in her experiment. Give your answer to the nearest **whole** number. [2]

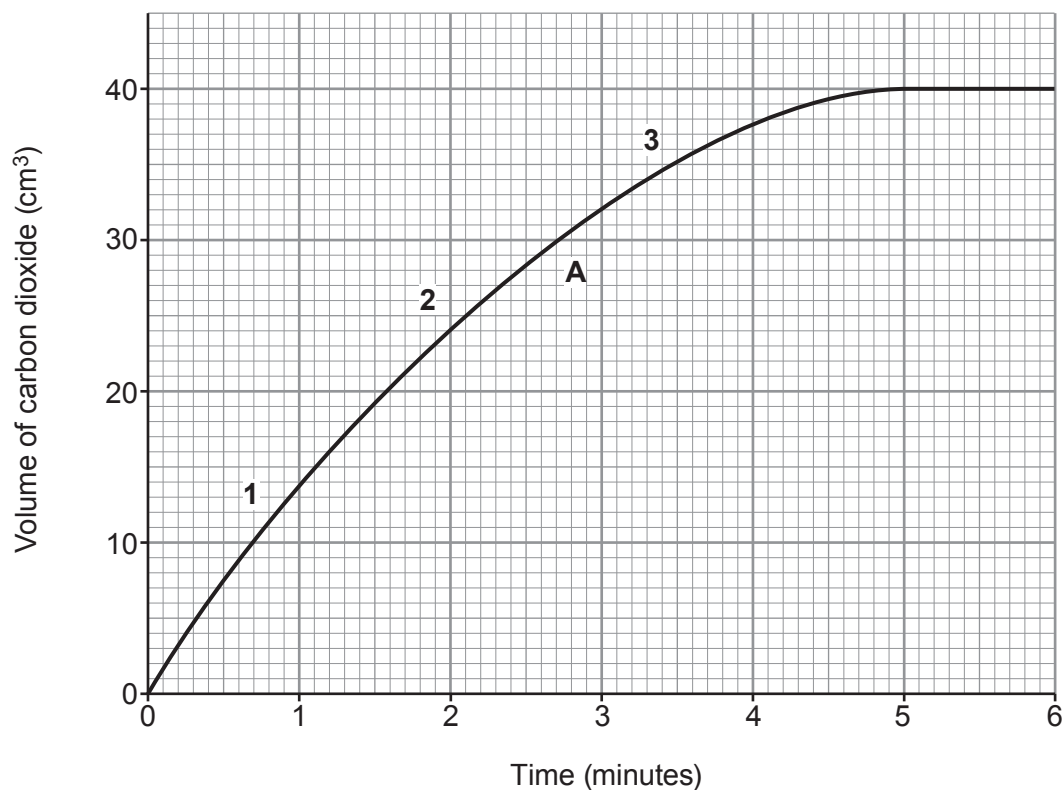
Percentage yield = %



4. Marble chips (calcium carbonate) react with hydrochloric acid to produce carbon dioxide gas as shown in the equation.



Graph **A** shows the volume of carbon dioxide produced over 6 minutes when the reaction was carried out at 30 °C.



- (a) (i) Find the volume of carbon dioxide given off after 2 minutes. [1]

..... cm³

- (ii) Find the time taken for the reaction to finish. [1]

..... minutes

- (iii) At which of the following points on graph **A** is the reaction at its fastest? [1]

Tick (✓) the correct box.

1

☐

2

☐

3

☐


- (iv) After $\frac{1}{2}$ minute, 7.5 cm^3 of carbon dioxide has been produced.

Use the equation to calculate the mean rate of reaction over this time in cm^3/s .

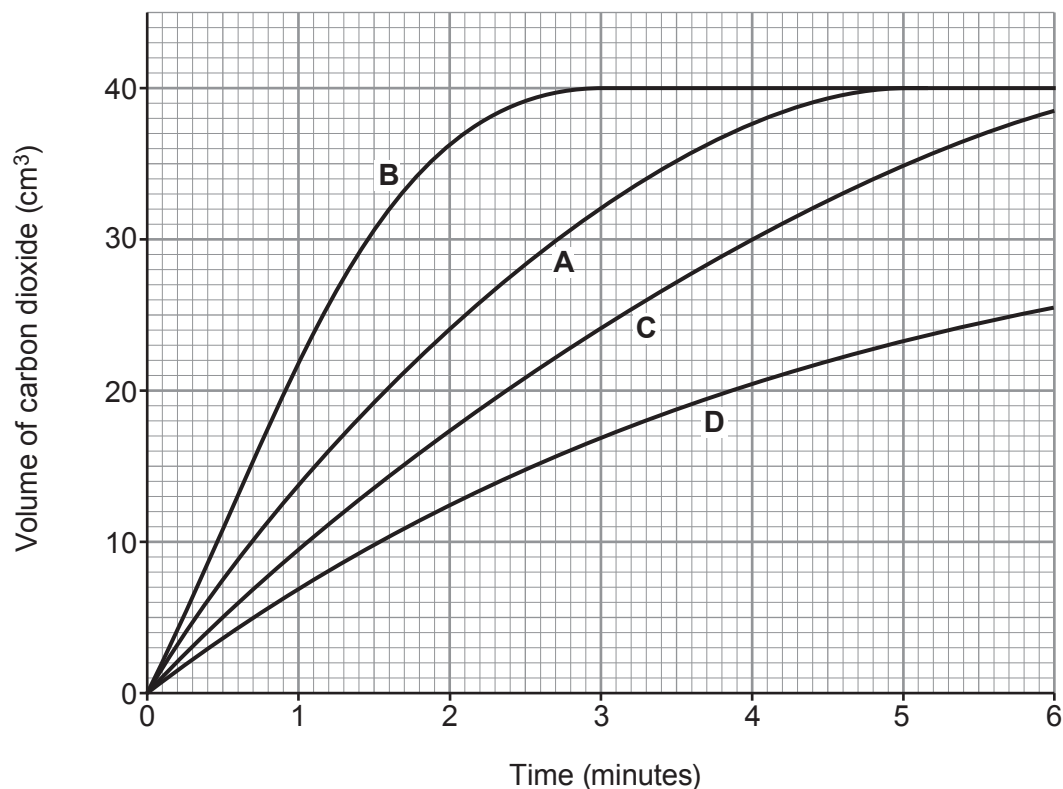
[2]

$$\text{mean rate} = \frac{\text{volume of carbon dioxide produced}}{\text{time (in seconds)}}$$

Mean rate = cm^3/s



- (b) (i) Graphs **B**, **C** and **D** on the grid below show the results for the experiment carried out at **different** temperatures. Graph **A** for the reaction carried out at 30 °C is also shown.



State which of graphs **B**, **C** and **D** shows the results for the experiment carried out at a temperature of 40 °C. [1]

Graph

- (ii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher temperature, the particles will have

(**more** / less / the same amount of) energy, so they will move faster and

there will be (an equal / a smaller / a greater) number of successful

collisions per second.



5. A teacher wanted to demonstrate the reactivity of lithium, sodium and potassium with water.

State what safety precautions the teacher should take and describe the observations you would expect for each metal.

You are **not** expected to include equations in your answer.

[6 QER]

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6. Fluoride ions (mostly from sodium fluoride) are found naturally in the drinking water in some regions of England. The concentration of these fluoride ions is quite low with values being less than 0.5 mg/dm^3 .

However, water companies in some areas add fluoride ions to drinking water. Areas where this happens include Yorkshire, the North East, the East Midlands and the West Midlands. The mean concentration of fluoride ions added is around 1 mg/dm^3 .

It is known that a high concentration of fluoride ions in drinking water can be poisonous and can cause a yellowing of teeth – fluorosis. Fluoridation of drinking water is a controversial issue. Some people are opposed to it because it removes the individual's choice to not take fluoride.

The table shows the mean number of decayed, missing and filled teeth (DMFT) in five-year-old children in some areas of England in 2018.

Area	Mean DMFT
South East	0.6
West Midlands	0.7
North West	1.3
London	1.0
South West	0.7
Yorkshire	1.0
North East	1.0
East Midlands	0.9
East England	0.7



- (a) Which **two** of the following statements are correct according to the information given? Tick (✓) **two** boxes. [2]

London, Yorkshire and the North West all have a mean DMFT of 1.0

☐

The South East has the lowest mean DMFT

☐

The East Midlands has the highest mean DMFT

☐

East England, the South West and the South East have a mean DMFT of 0.7

☐

The East Midlands has a lower mean DMFT than Yorkshire

☐

London, the North West and the South West all have a mean DMFT of 0.9

☐

- (b) (i) Which of the following statements supports the argument for fluoridation of drinking water? Tick (✓) **one** box. [1]

All areas with fluoridated drinking water have lower mean DMFT than areas which do not fluoridate drinking water

☐

All areas with fluoridated drinking water have the same mean DMFT as non-fluoridated areas

☐

All areas with fluoridated drinking water have lower mean DMFT than the North West

☐

- (ii) Which of the following statements could be used to oppose the argument for fluoridation of drinking water? Tick (✓) **one** box. [1]

It is not possible to decide whether fluoridation of drinking water lowers mean DMFT

☐

All areas with fluoridated drinking water have higher mean DMFT than areas which do not fluoridate drinking water

☐

All areas with fluoridated drinking water have a higher mean DMFT than the South West

☐


- (c) Suggest a possible reason why the South East has a low mean DMFT in five-year-old children even though no fluoride is added to drinking water in this area. [1]

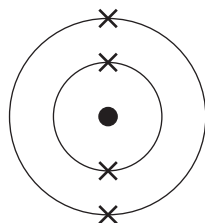
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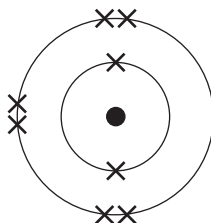


7. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.

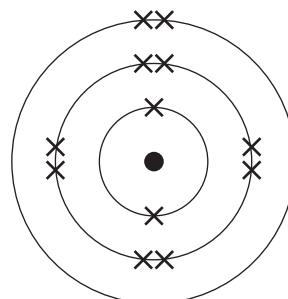
The letters are not the symbols of the elements.



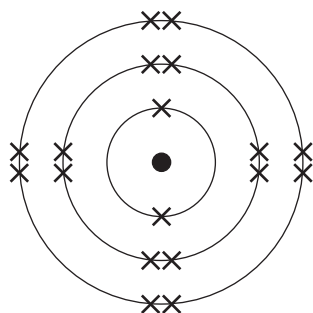
A



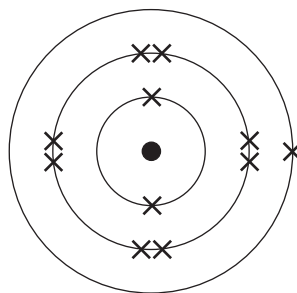
B



C



D



E

- (a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters and

Reason

.....



- (b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter

Reason

- (c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

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- (d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

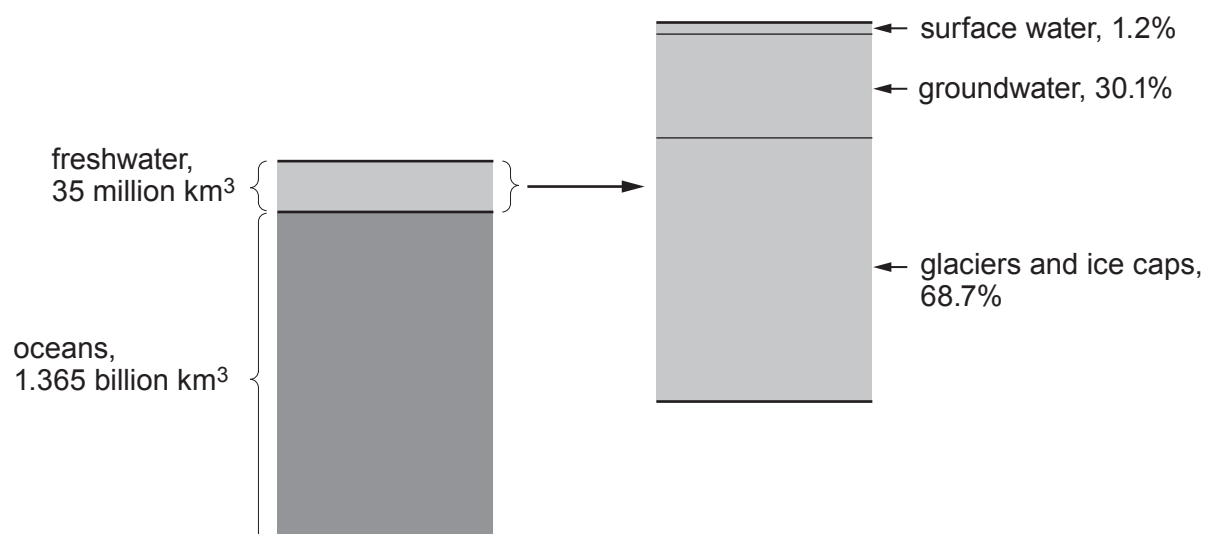
Give the formula for potassium oxide and balance the equation for this reaction. [2]



8



8. The diagram shows the distribution of water on Earth.



- (a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = million km³

- (b) As the Earth's population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination.

Give a method that could be used to obtain freshwater from seawater. [1]

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- (c) Calcium ions, Ca^{2+} , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

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- (d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

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END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

1 H Hydrogen 1 4 He Helium 2																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
										Key									

A_r	Symbol	Name	Z
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